QF620 Stochastic Modelling in Finance — Session 2 Reference Cheatsheet

Binomial Tree and the Risk-Neutral Measure: Radon–Nikodym derivative, arithmetic and geometric trees, American options, conditional expectation, martingales, risk-neutral pricing

Built from Session 2 slides (Tee Chyng Wen, SMU) · October 2026 · with symbol-by-symbol equation breakdowns

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Source: "Session 2: Binomial Tree and the Risk-Neutral Measure" (Tee Chyng Wen, SMU, 25 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. △**!** marks a slide ambiguity or error, with the corrected version.

How to read the equations: every displayed equation is followed by a shaded **where** box that names each symbol and component and says what it represents in that formula.

Exam context: the exam format is not stated on the Session 1–4 slides; check the course outline. From this session you need: the Radon–Nikodym derivative and how to reweight expectations; the no-arbitrage condition $d < 1+r < u$ and p^*; backward induction for European and American options on a small tree; the two CLT limits (normal and lognormal trees); conditional-expectation rules; and why discounted prices are $\mathbb{Q}$-martingales.

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
$\Omega,\ \omega$	sample space; one outcome (a price path)	set; element	[S2]
$\mathbb{P}$	real-world (empirical) measure	probability	e.g. fund manager's 80% up [S4]
$\mathbb{Q}$	risk-neutral measure	probability	also written with $*$: p^*, $\mathbb{E}^*$
$\frac{d\mathbb{Q}}{d\mathbb{P}}(\omega)$	Radon–Nikodym derivative	random variable	$=\mathbb{Q}(\omega)/\mathbb{P}(\omega)$ on a finite space [S2]
$S_n,\ S^u_{n+1},\ S^d_{n+1}$	stock at step n; up/down successors	price	
$u,\ d$	up and down factors	ratio	$u = S_1(H)/S_0$, $d = S_1(T)/S_0$ [S11]
r	risk-free rate per period (discrete)	rate	$\$1 \to \$(1+r)$ [S10]; △! continuous in [S15–S17]
$p^*,\ q^*$	risk-neutral up/down probabilities	probability	$p^* = \frac{(1+r)-d}{u-d}$, $q^* = 1-p^*$ [S19]
σ	volatility per $\sqrt{\text{time}}$	—	price units (arithmetic tree) or log units (geometric tree)
$T,\ n$ or N	maturity; number of steps	—	step size T/n
X_i	i.i.d. ±1 with probability $\frac{1}{2}$ each	random	drives the tree [S7, S15]
Z	standard normal	random	CLT limit [S8]
$V_n,\ V^E_n,\ V^A_n$	derivative value at step n; European; American	price	[S13, S23]
$\mathbb{E}^*_n[\cdot]$	conditional expectation under $\mathbb{Q}$ given the first n tosses	random	[S19]
M_n	discounted price $S_n/(1+r)^n$	random	a $\mathbb{Q}$-martingale [S21]
$(x)^+$	$\max(x, 0)$	—	payoff notation

Notation trap: [S10–S13, S19–S23] use a discrete, per-period rate $(1 + r)$; [S15–S17] use a continuously compounded rate (e^{rT}). The symbol r means different things in the two halves.

0. Topic map

```
PREREQUISITES
S1  Expectation, variance, MGF of the normal E[e^{θX}] = e^{μθ+σ²θ²/2}; lognormal E[e^X] = e^{μ+σ²/2};
    two-state hedging: Δ = (C_u − C_d)/(S_u − S_d); risk-neutral p* reproduces the hedge price
  |
  ▼
SESSION 2: BINOMIAL TREE AND THE RISK-NEUTRAL MEASURE
├─ RADON–NIKODYM  [S2–S6]
|   ├─ (Ω, P); dP/dQ(ω) = P(ω)/Q(ω)                                  [S2]
|   ├─ Needs equivalence: P(ω) > 0 ⇔ Q(ω) > 0                        [S3]
|   ├─ Example: P up 0.8, Q up 0.5, 100 ± 10 over two steps; dQ/dP tree   [S4]
|   └─ E^P[S₂] = 112, E^Q[S₂] = 100; E^Q[X] = E^P[X·dQ/dP]           [S5–S6]
|
├─ TREE MODELS  [S7–S17]
|   ├─ Arithmetic tree: ±σ√(T/n), prob ½ ⇒ S_T = S₀ + σ√T Z (CLT)    [S7–S9]
|   ├─ r ≠ 0: S_d < S₀(1+r) < S_u; p* = (S₀(1+r) − S_d)/(S_u − S_d)  [S10]
|   ├─ Geometric tree: u, d; p* = (1+r−d)/(u−d); CRR d = 1/u          [S11]
|   ├─ Example: European put, S₀=4, u=2, d=½, r=25%, K=5, 2 steps     [S12]
|   ├─ American: V = max(continuation, exercise)                      [S13]
|   ├─ CLT for products ⇒ lognormal                                   [S14]
|   └─ Log tree: ln S_T = ln S₀ + (r − ½σ²)T + σ√T Z; E[S_T] = S₀e^{rT}   [S15–S17]
|
├─ P AND Q  [S18–S20]
|   ├─ Coin example: random variables need no measure; distributions do   [S18]
|   ├─ E*_n[S_{n+1}] = p*S^u + q*S^d;  S_n = E*_n[S_{n+1}]/(1+r)      [S19]
|   └─ Linearity, taking out what is known, tower rule, independence  [S20]
|
└─ MARTINGALES & ARBITRAGE  [S21–S25]
    ├─ M_n = S_n/(1+r)^n is a Q-martingale; M₀ = E*[M_N]              [S21–S22]
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    ├─ Risk-neutral pricing formula V_n = E*_n[V_N/(1+r)^{N−n}]              [S23]
    └─ Complete markets, replication, no-arbitrage ⇒ Q exists; numeraires    [S24–S25]
  |
  ▼
LATER: S3 Brownian motion = limit of the ±√(t/n) random walk; continuous-time martingales;
       S4 Itô calculus; later Girsanov (continuous-time Radon–Nikodym) and Black–Scholes
```

Dependency logic:

- Session 1's one-period p^* becomes a whole probability measure $\mathbb{Q}$ on paths. The Radon–Nikodym derivative is the exchange rate between $\mathbb{P}$ and $\mathbb{Q}$ expectations.
- Backward induction with p^* at every node prices any payoff; American options add a max at each node.
- The CLT turns many small tree steps into a normal (arithmetic) or lognormal (geometric) terminal distribution: the bridge to Brownian motion (S3) and Black–Scholes.
- Conditional-expectation rules show that discounted prices are martingales under $\mathbb{Q}$, which is exactly the risk-neutral pricing formula.

1 Concept entries

1.1 Probability spaces and the Radon–Nikodym derivative [S2–S3]

- **Probability space** [S2]: $(\Omega, \mathbb{P})$. Ω lists what can happen; $\mathbb{P}$ decides how likely each outcome is.
- **Changing measure** [S2]: keeps the same paths but changes their likelihoods.
- **Radon–Nikodym derivative** [S2] (finite Ω):

$$\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}(\omega) = \frac{\mathbb{P}(\omega)}{\mathbb{Q}(\omega)}, \qquad \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}(\omega) = \frac{\mathbb{Q}(\omega)}{\mathbb{P}(\omega)}.$$

where

Ω	sample space: the set of all possible outcomes (in a tree, all coin-toss paths, e.g. UU, UD, DU, DD).
ω	one specific outcome/path, $\omega \in \Omega$.
$\mathbb{P}(\omega)$	probability of path ω under the real-world measure $\mathbb{P}$ (product of the per-step probabilities along the path).
$\mathbb{Q}(\omega)$	probability of the same path under the risk-neutral measure $\mathbb{Q}$.
$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}(\omega)$	the Radon–Nikodym derivative: a *random variable* (one number per path) equal to the ratio of the two probabilities; it is the weight that converts $\mathbb{P}$-probabilities into $\mathbb{Q}$-probabilities.
$\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}(\omega)$	the reverse weight, the reciprocal of $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$. The "d" is notation, not a derivative in the calculus sense on a finite space.

- **Equivalence** [S3]: $\mathbb{P}$ and $\mathbb{Q}$ are equivalent if they live on the same Ω and agree on what is possible:

$$\mathbb{P}(\omega) > 0 \iff \mathbb{Q}(\omega) > 0.$$

where

$\mathbb{P}(\omega) > 0$	path ω is possible under the real-world measure.
$\mathbb{Q}(\omega) > 0$	the same path is possible under the risk-neutral measure.
$\iff$	"if and only if": both measures must agree on which paths are possible and which are impossible. Only then are both ratios defined on every path.

- **Why it matters** [S3]: dropping outcomes that are $\mathbb{P}$-impossible but $\mathbb{Q}$-possible loses information about $\mathbb{Q}$. Equivalence is the formal version of Session 1's "only zero vs non-zero probabilities matter".
- **Properties** *[Beyond slides]*:

$$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} > 0, \qquad \mathbb{E}^{\mathbb{P}}\left[\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right] = \sum_{\omega} \mathbb{Q}(\omega) = 1, \qquad \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}} = \left(\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right)^{-1}.$$

where

$\mathbb{E}^{\mathbb{P}}[\cdot]$	expectation computed with $\mathbb{P}$-probabilities: $\sum_{\omega}(\cdot)\,\mathbb{P}(\omega)$.
$\sum_{\omega} \mathbb{Q}(\omega) = 1$	because $\mathbb{P}(\omega) \cdot \frac{\mathbb{Q}(\omega)}{\mathbb{P}(\omega)} = \mathbb{Q}(\omega)$, summing over every path gives total $\mathbb{Q}$-probability, which is 1. This is a quick check that your weights are right.
$(\cdot)^{-1}$	reciprocal; the two derivatives undo each other.

- △! [S3] says that if $\mathbb{P}(\omega_i) = 0$ but $\mathbb{Q}(\omega_i) \neq 0$ then $\mathbb{P}(\omega_i)/\mathbb{Q}(\omega_i)$ is undefined. That ratio is 0 (defined); it is $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P} = \mathbb{Q}/\mathbb{P}$ that breaks. The conclusion (you need equivalence for both directions) is right.
- △! [S2] A probability space is properly a triple $(\Omega, \mathcal{F}, \mathbb{P})$ with a set of events $\mathcal{F}$; the slides drop $\mathcal{F}$ because Ω is finite.

1.2 Radon–Nikodym example: bullish fund manager vs market maker [S4–S6]

- **Setup** [S4]: $S_0 = 100$, $\pm\$10$ per period, $r = 0$, two periods. Fund manager: $\mathbb{P}(\text{up}) = 0.8$. Market maker: risk-neutral $\mathbb{Q}(\text{up}) = 0.5$.
- **Radon–Nikodym process** $d\mathbb{Q}/d\mathbb{P}$ [S4] (multiply the per-step ratios along the path):

Node	$t=0$	$t=1$	$t=2$
up-up ($S=120$)		up: $0.5/0.8 = 5/8$	$25/64$
middle ($S=100$)	1		$(5/8)(5/2) = 25/16$
down-down ($S=80$)		down: $0.5/0.2 = 5/2$	$25/4$

Expectations [S5]:

$$\mathbb{E}^{\mathbb{P}}[S_2] = 0.8^2(120) + 2(0.8)(0.2)(100) + 0.2^2(80) = 76.8 + 32 + 3.2 = 112.$$

where

S_2	stock price at the end of period 2 (one of 120, 100, 80).
0.8^2	$\mathbb{P}$-probability of up-up (two independent ups).
$2(0.8)(0.2)$	$\mathbb{P}$-probability of ending in the middle: two paths (UD and DU), each with probability 0.8×0.2.
0.2^2	$\mathbb{P}$-probability of down-down.
112	the fund manager's forecast of S_2: a *view*, not a price.

$$\mathbb{E}^{\mathbb{Q}}[S_2] = 0.5^2(120) + 2(0.5)(0.5)(100) + 0.5^2(80) = 100.$$

where

$0.5^2,\ 2(0.5)(0.5)$	the same path counts with risk-neutral probabilities: 0.25, 0.5, 0.25.
100	equals S_0 (since $r = 0$): under $\mathbb{Q}$ the stock has no expected gain.

- **Enforceable vs not** [S5]: the market maker can lock in 100 (borrow 100, buy the stock, owe exactly 100 at $t = 2$). The manager's 112 carries risk and cannot be enforced.
- **Change-of-measure formulas** [S5–S6]:

$$\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}\left[X\,\frac{d\mathbb{Q}}{d\mathbb{P}}\right], \qquad \mathbb{E}^{\mathbb{P}}[X] = \mathbb{E}^{\mathbb{Q}}\left[X\,\frac{d\mathbb{P}}{d\mathbb{Q}}\right].$$

where

X	any random variable defined on the paths (here $X = S_2$, but it can be any payoff).
$\mathbb{E}^{\mathbb{Q}}[X]$	expectation of X using $\mathbb{Q}$-probabilities.
$X\frac{d\mathbb{Q}}{d\mathbb{P}}$	X reweighted path by path; averaging this with $\mathbb{P}$-probabilities gives the $\mathbb{Q}$-average because $\mathbb{P}(\omega)\frac{\mathbb{Q}(\omega)}{\mathbb{P}(\omega)} = \mathbb{Q}(\omega)$.
$\frac{d\mathbb{P}}{d\mathbb{Q}}$	the reverse weight, used to go from a $\mathbb{Q}$-average back to a $\mathbb{P}$-average.

Check [S6]:

$$0.64(120)\tfrac{25}{64} + 0.32(100)\tfrac{25}{16} + 0.04(80)\tfrac{25}{4} = 30 + 50 + 20 = 100\ \checkmark.$$

where

0.64, 0.32, 0.04	$\mathbb{P}$-probabilities of the three terminal nodes.
120, 100, 80	terminal stock prices S_2.
$\frac{25}{64},\ \frac{25}{16},\ \frac{25}{4}$	$d\mathbb{Q}/d\mathbb{P}$ at those nodes.

- **Interpretation** [S6]: $d\mathbb{Q}/d\mathbb{P} < 1$ on high prices and > 1 on low prices: it moves weight toward the low outcomes, bringing the expectation down to the risk-neutral level.
- *[Beyond slides]* Sanity check $\mathbb{E}^{\mathbb{P}}[d\mathbb{Q}/d\mathbb{P}] = 0.64(25/64) + 0.32(25/16) + 0.04(25/4) = 0.25 + 0.5 + 0.25 = 1$ ✓. The process $Z_n = \mathbb{E}^{\mathbb{P}}_n[d\mathbb{Q}/d\mathbb{P}]$ (the tree values) is a $\mathbb{P}$-martingale.
- **Traps:** inverting the ratio ($d\mathbb{Q}/d\mathbb{P} = \mathbb{Q}/\mathbb{P}$, not $\mathbb{P}/\mathbb{Q}$); computing it per step but forgetting to multiply along the path.

1.3 Arithmetic (normal) binomial tree and the CLT [S7–S9]

Goal [S7] ($r = 0$):

$$\mathbb{E}[S_T] = S_0, \qquad \mathbb{V}[S_T] = \sigma^2 T.$$

where

S_0	today's stock price.
S_T	stock price at maturity T (random).
$\mathbb{E}[S_T] = S_0$	no drift: with $r = 0$ the expected future price equals today's price.
$\mathbb{V}[S_T]$	variance of the terminal price.
σ	volatility, here in *price units* per $\sqrt{\text{time}}$ (an absolute, not percentage, volatility).
T	time to maturity; variance grows linearly in T.

Construction [S7]: n steps of length T/n; each step has mean 0 and variance $\sigma^2 T/n$, so the stock moves $\pm\sigma\sqrt{T/n}$ with probability $\frac{1}{2}$:

$$S_T = S_0 + \sum_{i=1}^{n} \sigma\sqrt{\frac{T}{n}}\, X_i, \qquad X_i = \pm 1.$$

where

n	number of time steps; each step has length $\Delta t = T/n$.
X_i	independent coin-toss variable for step i: $+1$ (up) or -1 (down), each with probability $\frac{1}{2}$; mean 0, variance 1.
$\sigma\sqrt{T/n}$	size of one move. Squared it gives the per-step variance $\sigma^2 T/n$, so n steps add up to $\sigma^2 T$.
$\sum_{i=1}^{n}$	total displacement after n steps (a random walk).

Central Limit Theorem [S8]:

$$Z = \frac{Y_n - n\mu}{\sqrt{ns^2}} \xrightarrow{n\to\infty} N(0,1), \qquad Y_n = X_1 + X_2 + \cdots + X_n.$$

where

$X_1, X_2, \ldots$	i.i.d. (independent, identically distributed) random variables.
μ	common mean of each X_i (here 0).
s^2	common variance of each X_i (here 1).
Y_n	the sum of the first n variables.
$n\mu,\ \sqrt{ns^2}$	mean and standard deviation of Y_n; subtracting and dividing *standardises* the sum.
$N(0,1)$	standard normal distribution (mean 0, variance 1).

With $\mu = 0,\ s^2 = 1$:

$$\lim_{n\to\infty} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \sim N(0,1) \quad \Longrightarrow \quad S_T = S_0 + \sigma\sqrt{T}\, Z, \qquad Z \sim N(0,1).$$

where

$\frac{1}{\sqrt{n}}\sum X_i$	the standardised random walk; factor out $\sigma\sqrt{T/n} = \sigma\sqrt{T}\cdot\frac{1}{\sqrt{n}}$ from the tree sum to see it.
Z	standard normal random variable: the limit of the standardised sum.
$\sigma\sqrt{T}\,Z$	normal shock with mean 0 and standard deviation $\sigma\sqrt{T}$.
S_T	normally distributed: $S_T \sim N(S_0, \sigma^2 T)$ (the Bachelier model).

- **Slide example** [S9]: $\mathbb{E}[S_T] = S_0 + \sigma\sqrt{T}\,\mathbb{E}[Z] = S_0$; $\mathbb{V}[S_T] = \sigma^2 T\,\mathbb{V}[Z] = \sigma^2 T$ ✓ (constants come out of $\mathbb{E}$ unchanged and out of $\mathbb{V}$ squared).
- **Why** $\sqrt{T/n}$ *[Beyond slides]*: the step must shrink like $\sqrt{\Delta t}$ so the variance adds up to $\sigma^2 T$. Steps of Δt would give zero variance; steps of size 1 would explode. This is exactly the scaling of Brownian motion (S3).
- **Weakness** *[Beyond slides]*: S_T can be negative.

1.4 Non-zero interest rates and the no-arbitrage condition [S10]

Rate [S10]: \$1 today grows to $\$(1+r)$ next period. **No-arbitrage condition** [S10]:

$$S_1^d < S_0(1+r) < S_1^u \qquad \text{(equivalently } d < 1+r < u\text{)}.$$

where

S_0	stock price today.
$S_1^u,\ S_1^d$	stock price next period after an up move / down move.
r	risk-free rate per period (discrete, simple).
$S_0(1+r)$	what you get by putting S_0 in the bank instead of buying the stock.
inequality	the bank outcome must sit strictly between the two stock outcomes; otherwise one asset beats the other in every state (an arbitrage).

- *[Beyond slides]* If $S_0(1+r) \le S_1^d$: borrow S_0, buy the stock; you end with at least what you owe, strictly more in the up state.
- *[Beyond slides]* If $S_0(1+r) \ge S_1^u$: short the stock, invest the proceeds.

Risk-neutral probability [S10]: require the stock to grow at the risk-free rate:

$$\mathbb{E}^*[S_1] = p^* S_1^u + (1-p^*)S_1^d = S_0(1+r) \quad \Longrightarrow \quad p^* = \frac{S_0(1+r) - S_1^d}{S_1^u - S_1^d} \in (0,1) \iff \text{no arbitrage.}$$

where

$\mathbb{E}^*[\cdot]$	expectation under the risk-neutral measure $\mathbb{Q}$.
p^*	risk-neutral probability of the up move (not a real-world belief; it is implied by prices).
$1-p^* = q^*$	risk-neutral probability of the down move.
$S_0(1+r) - S_1^d$	how far the risk-free outcome sits above the down outcome (numerator).
$S_1^u - S_1^d$	total spread between up and down outcomes (denominator).
$\in (0,1)$	p^* is a valid probability exactly when the no-arbitrage inequality holds.

Pricing *[Beyond slides]*:

$$V_0 = \frac{p^* V_1^u + q^* V_1^d}{1+r}, \qquad \Delta = \frac{V^u - V^d}{S^u - S^d}, \qquad B = \frac{uV^d - dV^u}{(u-d)(1+r)}, \qquad V_0 = \Delta S_0 + B.$$

where

V_0	derivative price today.
$V_1^u,\ V_1^d$	derivative value next period in the up / down state (payoffs at expiry).
$\dfrac{1}{1+r}$	one-period discount factor.
Δ	hedge ratio: number of shares to hold so that the portfolio moves exactly like the derivative.
B	amount in the bank (negative = borrowing).
$\Delta S_0 + B$	cost of the replicating portfolio; equals the risk-neutral price.

Trap: forgetting to discount by $(1+r)$; using $\mathbb{E}^*[S_1] = S_0$ when $r \neq 0$.

1.5 Geometric (lognormal) binomial tree [S11–S12]

Multiplicative moves [S11]:

$$S_1^u = u\,S_0, \quad S_1^d = d\,S_0, \qquad u = \frac{S_1(H)}{S_0}, \quad d = \frac{S_1(T)}{S_0}.$$

where

u	up factor (gross return in the up state), e.g. $u=2$ doubles the price.
d	down factor (gross return in the down state), $0 < d < u$.
$S_1(H),\ S_1(T)$	next-period price if the coin shows heads (up) / tails (down). Not to be confused with maturity T.

Risk-neutral probability [S11]:

$$p^* = \frac{(1+r) - d}{u - d}, \qquad q^* = \frac{u - (1+r)}{u - d}.$$

where

$1+r$	gross risk-free return per period.
$(1+r) - d$	how much the bank beats the down move.
$u - (1+r)$	how much the up move beats the bank.
$u - d$	spread between the two gross returns; makes $p^* + q^* = 1$.

It does not depend on S_0, so it is the same at every node. **Cox–Ross–Rubinstein** [S11]: choose $d = 1/u$ (tree recombines symmetrically in log space). *[Beyond slides]* Usual calibration $u = e^{\sigma\sqrt{\Delta t}}$, $\Delta t = T/N$.

Slide example [S12]: $S_0 = 4$, $u = 2$, $d = \frac{1}{2}$, $r = 25\%$, European put $K = 5$, expiry $t = 2$.

1. $p^* = (1.25 - 0.5)/(2 - 0.5) = 0.5$, $q^* = 0.5$.
2. Terminal prices $16, 4, 1$; put payoffs $(K - S_2)^+ = 0, 1, 4$.
3. $t = 1$: $V^u(S = 8) = (0.5 \cdot 0 + 0.5 \cdot 1)/1.25 = 0.40$; $V^d(S = 2) = (0.5 \cdot 1 + 0.5 \cdot 4)/1.25 = 2.00$.
4. $t = 0$: $V_0 = (0.5 \cdot 0.40 + 0.5 \cdot 2.00)/1.25 = 0.96$.

[Beyond slides] Direct (one-shot) formula:

$$V_0 = \frac{1}{(1+r)^N} \sum_{j=0}^{N} \binom{N}{j} p^{*j} q^{*\,N-j}\, V_N(j) = \frac{1}{1.25^2}\left[0.25(0) + 0.5(1) + 0.25(4)\right] = \frac{1.5}{1.5625} = 0.96 \checkmark$$

where

N	number of periods to expiry (here 2).
j	number of up moves on the path; terminal price $S_0 u^j d^{N-j}$.
$\binom{N}{j}$	number of paths with exactly j ups (the binomial coefficient).
$p^{*j} q^{*\,N-j}$	risk-neutral probability of one such path.
$V_N(j)$	payoff at expiry in the node with j ups.
$\frac{1}{(1+r)^N}$	discount factor over all N periods.

The call with $K = 5$ is $0.25(11)/1.5625 = 1.76$. Put–call parity:

$$C - P = S_0 - \frac{K}{(1+r)^N}: \qquad 1.76 - 0.96 = 0.80 = 4 - \frac{5}{1.5625} \checkmark$$

where

C, P	European call and put prices (same strike and expiry).
S_0	stock price today.
$K/(1+r)^N$	present value of the strike.

Implementation *[Beyond slides]* (tested):

```
import numpy as np
def binomial_price(S0, K, u, d, r, N, payoff="put", american=False):
    """N-step recombining tree; r = per-period simple rate (1+r per step)."""
    p = ((1 + r) - d) / (u - d)
    assert 0 < p < 1, "no-arbitrage requires d < 1+r < u"
    j = np.arange(N + 1)
    ex = (lambda s: np.maximum(K - s, 0)) if payoff == "put" else (lambda s: np.maximum(s - K, 0))
    V = ex(S0 * u**j * d**(N - j))                      # terminal payoffs, j = # ups
    for n in range(N - 1, -1, -1):
        V = (p * V[1:] + (1 - p) * V[:-1]) / (1 + r)    # continuation value
        if american:
            V = np.maximum(V, ex(S0 * u**j[:n+1] * d**(n - j[:n+1])))
    return V[0]
print(binomial_price(4, 5, 2, 0.5, 0.25, 2))                # 0.96
print(binomial_price(4, 5, 2, 0.5, 0.25, 2, american=True))  # 1.36
```

1.6 American options [S13]

Rule [S13]: exercise only if it pays more than continuing.

$$V_n^E = \frac{1}{1+r}\mathbb{E}_n^*[V_{n+1}] = \frac{p^* V_{n+1}^u + q^* V_{n+1}^d}{1+r}, \qquad V_n^A = \max\left(\frac{p^* V_{n+1}^u + q^* V_{n+1}^d}{1+r},\ (K - S_n)^+\right) \text{ (put).}$$

where

V_n^E	European option value at a node at step n.
V_n^A	American option value at that node.
V_{n+1}^u, V_{n+1}^d	option values at the two successor nodes (for an American, use the *American* values there).
$\frac{p^* V^u + q^* V^d}{1+r}$	**continuation value**: what the option is worth if you keep holding it one more step.
$(K - S_n)^+$	**exercise (intrinsic) value** of a put: $\max(K - S_n, 0)$, what you get by exercising now. For a call use $(S_n - K)^+$.
K	strike price.
$\max(\cdot, \cdot)$	the holder picks the better of continuing and exercising, at every node.

Slide example [S13]: American put on the [S12] tree.

- $S = 8$: continuation 0.40, exercise $(5-8)^+ = 0 \Rightarrow 0.40$.
- $S = 2$: continuation 2.00, exercise $5 - 2 = 3 \Rightarrow 3.00$ (exercise early).
- $t = 0$: continuation $(0.5 \cdot 0.40 + 0.5 \cdot 3.00)/1.25 = 1.36$; exercise $5 - 4 = 1 \Rightarrow V_0^A = 1.36$.
- Early-exercise premium $= V_0^A - V_0^E = 1.36 - 0.96 = 0.40$.
- *[Beyond slides]* An American call on a non-dividend stock with $r > 0$ is never exercised early (its value equals the European call); American puts can be.
- **Traps:** comparing with the exercise value only at $t = 0$ (you must check every node); using the European values at $t = 1$ when rolling back.

1.7 CLT for products: the lognormal tree [S14–S17]

CLT for products [S14]:

$$W = X_1 \times X_2 \times \cdots \times X_n, \quad X_i \in (0, \infty) \quad \Longrightarrow \quad \ln W = \ln X_1 + \ln X_2 + \cdots + \ln X_n.$$

where

X_i	independent positive random factors (e.g. gross returns u or d per step).
W	their product (e.g. S_T/S_0).
$\ln W$	a *sum* of independent terms $\ln X_i$; if each has finite mean and variance the CLT makes $\ln W$ approximately normal, so W is approximately **lognormal**.
$X_i > 0$	needed so that $\ln X_i$ exists.

Targets [S15] (risk-neutral, r continuous):

$$\mathbb{E}[\ln S_T] = \ln S_0 + \left(r - \tfrac{\sigma^2}{2}\right) T, \qquad \mathbb{V}[\ln S_T] = \sigma^2 T.$$

where

$\ln S_T$	log of the terminal price (the log-price).
r	continuously compounded risk-free rate (△**!** not the per-period simple rate of [S10–S13]).
σ	volatility of log returns (percentage volatility) per $\sqrt{\text{time}}$.
$r - \frac{1}{2}\sigma^2$	drift of the log-price; the $-\frac{1}{2}\sigma^2$ is the convexity correction.
$\sigma^2 T$	variance of the log-price, growing linearly in time.

Log tree [S15]:

$$\ln S_i = \ln S_{i-1} + \left(r - \tfrac{1}{2}\sigma^2\right)\frac{T}{n} + \sigma\sqrt{\frac{T}{n}}\,X_i \;\Longrightarrow\; \ln S_T = \ln S_0 + \left(r - \tfrac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\,\frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i.$$

where

$\ln S_i,\ \ln S_{i-1}$	log-price after step i and before it.
$(r - \frac{1}{2}\sigma^2)\frac{T}{n}$	deterministic drift per step (step length T/n).
$\sigma\sqrt{T/n}\,X_i$	random log-shock per step, $X_i = \pm 1$ with probability $\frac{1}{2}$.
summing $i = 1..n$	the drift terms add to $(r - \frac{1}{2}\sigma^2)T$; the shocks add to $\sigma\sqrt{T} \cdot \frac{1}{\sqrt{n}} \sum X_i$.

Limit [S16] (CLT applied to $\frac{1}{\sqrt{n}} \sum X_i \to Z$):

$$\ln S_T = \ln S_0 + \left(r - \tfrac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\,Z \quad \Longrightarrow \quad S_T = S_0 \exp\Big[\left(r - \tfrac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\,Z\Big], \qquad Z \sim N(0,1).$$

where

Z	standard normal shock.
$\exp[\cdot]$	exponentiating the log-price returns the price; $S_T > 0$ always.
S_T	lognormal: $\ln S_T \sim N\big(\ln S_0 + (r - \frac{1}{2}\sigma^2)T,\ \sigma^2 T\big)$. This is the risk-neutral geometric Brownian motion / Black–Scholes terminal distribution.

Slide example [S16]: $\mathbb{E}[\ln S_T] = \ln S_0 + (r - \frac{1}{2}\sigma^2)T$, $\mathbb{V}[\ln S_T] = \sigma^2 T$ ✓ (mean and variance of an affine function of Z).

Slide example [S17] (MGF of Z, $\mathbb{E}[e^{\theta Z}] = e^{\theta^2/2}$, at $\theta = \sigma\sqrt{T}$):

$$\mathbb{E}[S_T] = S_0 e^{(r-\frac{1}{2}\sigma^2)T}\,\mathbb{E}\big[e^{\sigma\sqrt{T}Z}\big] = S_0 e^{(r-\frac{1}{2}\sigma^2)T} e^{\frac{1}{2}\sigma^2 T} = S_0 e^{rT}.$$

where

$S_0 e^{(r-\frac{1}{2}\sigma^2)T}$	the non-random part, pulled out of the expectation.
$\mathbb{E}[e^{\theta Z}] = e^{\theta^2/2}$	moment generating function of a standard normal.
$\theta = \sigma\sqrt{T}$	so the MGF gives $e^{\frac{1}{2}\sigma^2 T}$, which cancels the $-\frac{1}{2}\sigma^2 T$.
$S_0 e^{rT}$	the stock grows at the risk-free rate on average: the no-arbitrage requirement under $\mathbb{Q}$.

- **Why** $-\frac{1}{2}\sigma^2$ *[Beyond slides]*: it cancels the lognormal convexity $e^{\frac{1}{2}\sigma^2 T}$ (S1). Without it, $\mathbb{E}[S_T] = S_0 e^{(r+\frac{1}{2}\sigma^2)T}$, violating no-arbitrage.
- **Convergence to Black–Scholes** *[Beyond slides]*: CRR with $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $1 + r = e^{r\Delta t}$; call $S_0 = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$: $N = 10$: 10.2534; $N = 100$: 10.4306; $N = 1000$: 10.4486; Black–Scholes 10.4506.
- △! [S15] The $\pm\sigma\sqrt{T/n}$ log tree with probability $\frac{1}{2}$ is only risk-neutral in the limit. One step gives *[Beyond slides]*:

$$\mathbb{E}\left[\frac{S_i}{S_{i-1}}\right] = e^{(r-\frac{1}{2}\sigma^2)\Delta t}\cosh\big(\sigma\sqrt{\Delta t}\big) = e^{r\Delta t}\big(1 + O(\Delta t^2)\big).$$

where

S_i/S_{i-1}	one-step gross return.
$\Delta t = T/n$	step length.
$\cosh(x) = \frac{1}{2}(e^x + e^{-x})$	average of the up and down log-shock factors with equal probability.
$O(\Delta t^2)$	error term of order Δt^2; vanishes in the limit, so the tree is risk-neutral only asymptotically.

1.8 Random variables vs measures: the coin example [S18]

- **Setup** [S18]: toss a coin 3 times; X = number of heads, Y = number of tails.
- **No measure needed to define them** [S18]: they are functions on $\Omega = \{HHH, \ldots, TTT\}$, e.g. $X(HHT) = 2$, $Y(HHT) = 1$; always $Y = 3 - X$ *[Beyond slides]*.
- **Distributions do need a measure:**
  - Fair coin: $X, Y \sim \text{Bin}(3, \frac{1}{2})$, same distribution (probabilities $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$).
  - $p = \frac{2}{3}$ *[Beyond slides]*: $\mathbb{P}(X = 0, 1, 2, 3) = \frac{1}{27}, \frac{6}{27}, \frac{12}{27}, \frac{8}{27}$; Y has these reversed. $\mathbb{E}[X] = 2$, $\mathbb{E}[Y] = 1$.
- **Expectation on a finite space** [S18]:

$$\mathbb{E}[X] = \sum_{\omega\in\Omega} X(\omega)\,\mathbb{P}(\omega).$$

where

Ω	all 8 toss sequences.
$X(\omega)$	value of the random variable on outcome ω (fixed, measure-free).
$\mathbb{P}(\omega)$	probability of that outcome (this is where the measure enters).
$\sum_{\omega\in\Omega}$	probability-weighted average over every outcome.

Lesson: changing the measure changes distributions and expectations, never the random variables (payoffs) themselves.

1.9 Conditional expectation [S19–S20]

One-step pricing as a conditional expectation [S19]:

$$\mathbb{E}^*_n[S_{n+1}] = p^* S^u_{n+1} + q^* S^d_{n+1}, \qquad S_n = \frac{1}{1+r}\,\mathbb{E}^*_n[S_{n+1}].$$

where

$\mathbb{E}^*_n[\cdot]$	expectation under $\mathbb{Q}$ given the information at t_n (the first n tosses, i.e. which node you are at). It is itself a random variable: its value depends on the node.
S_{n+1}	next-step price, $S^u_{n+1} = uS_n$ or $S^d_{n+1} = dS_n$.
p^*, q^*	risk-neutral up/down probabilities $\frac{(1+r)-d}{u-d}$, $\frac{u-(1+r)}{u-d}$.
$\frac{1}{1+r}$	one-period discount factor; today's price is the discounted risk-neutral forecast of tomorrow's.

Properties [S20]:

Linearity:	$\mathbb{E}_n[aX + bY] = a\mathbb{E}_n[X] + b\mathbb{E}_n[Y]$
Taking out what is known:	$\mathbb{E}_n[XY] = X\,\mathbb{E}_n[Y]$ if X depends only on the first n tosses
Iterated expectation (tower):	$\mathbb{E}_n\big[\mathbb{E}_m[X]\big] = \mathbb{E}_n[X], \quad 0 \le n \le m \le N$
Independence:	$\mathbb{E}_n[X] = \mathbb{E}[X]$ if X depends only on toss $n+1$

where

$a,\ b$	constants.
$X,\ Y$	random variables (payoffs/prices on the tree).
$\mathbb{E}_n$	conditional expectation given the first n tosses.
$\mathbb{E}_m$	conditional expectation at a later time $m \ge n$.
"known at n"	X is fixed once you know the first n tosses, so it behaves like a constant inside $\mathbb{E}_n$.
tower	the smaller information set wins: averaging a later forecast with today's information gives today's forecast.
independence	information about tosses $1..n$ says nothing about toss $n+1$, so the conditional and unconditional expectations agree.
N	final time step.

- **Tip** *[Beyond slides]*: every martingale proof in S2–S4 is these four rules plus "increments are independent".
- **Trap:** using the tower rule in the wrong order: $\mathbb{E}_m[\mathbb{E}_n[X]] = \mathbb{E}_n[X]$ too (for $n \le m$, $\mathbb{E}_n[X]$ is already known at m), but $\mathbb{E}_n[\mathbb{E}_m[X]]$ collapses to the earlier time.

1.10 Martingales and the risk-neutral pricing formula [S21–S23]

Discounted stock is a $\mathbb{Q}$-martingale [S21]: divide $S_n = \frac{1}{1+r}\mathbb{E}^*_n[S_{n+1}]$ by $(1+r)^n$:

$$\frac{S_n}{(1+r)^n} = \mathbb{E}^*_n\left[\frac{S_{n+1}}{(1+r)^{n+1}}\right], \qquad M_n = \mathbb{E}^*_n[M_{n+1}], \quad n = 0, 1, \ldots, N-1, \qquad M_0 = \mathbb{E}^*[M_N].$$

where

$(1+r)^n$	value at t_n of \$1 deposited in the bank at t_0 (the bank account, the numeraire).
$M_n = \dfrac{S_n}{(1+r)^n}$	discounted stock price: the stock measured in units of the bank account.
$M_n = \mathbb{E}^*_n[M_{n+1}]$	**martingale property**: the best $\mathbb{Q}$-forecast of tomorrow's discounted price is today's.
$M_0 = \mathbb{E}^*[M_N]$	applying the tower rule repeatedly: today's price equals the risk-neutral expectation of the discounted terminal price.
$\mathbb{E}^* = \mathbb{E}^*_0$	unconditional (time-0) risk-neutral expectation.

Proof [S22] (needs $0 < d < 1 + r < u$):

$$\mathbb{E}^*_n\left[\frac{S_{n+1}}{(1+r)^{n+1}}\right] = \frac{S_n}{(1+r)^n}\cdot\frac{1}{1+r}\,\mathbb{E}^*_n\left[\frac{S_{n+1}}{S_n}\right] = \frac{S_n}{(1+r)^n}\cdot\frac{p^*u + q^*d}{1+r} = \frac{S_n}{(1+r)^n}.$$

where

$\dfrac{S_{n+1}}{S_n}$	one-step gross return: u or d, depends only on toss $n+1$.
step 1	*taking out what is known*: S_n is known at t_n, so it leaves $\mathbb{E}^*_n$.
step 2	*independence*: $\mathbb{E}^*_n[S_{n+1}/S_n] = p^*u + q^*d$.
$p^*u + q^*d = 1 + r$	*[Beyond slides]* plug in p^*: $\frac{(1+r-d)u+(u-1-r)d}{u-d} = 1+r$; the risk-neutral expected gross return is the risk-free one, so the last fraction equals 1.

Risk-neutral pricing formula [S23]: for a payoff V_N at t_N, the discounted derivative price is a $\mathbb{Q}$-martingale and

$$\frac{V_n}{(1+r)^n} = \mathbb{E}^*_n\left[\frac{V_{n+1}}{(1+r)^{n+1}}\right], \qquad V_n = \mathbb{E}^*_n\left[\frac{V_N}{(1+r)^{N-n}}\right], \quad 0 \le n \le N.$$

where

V_N	derivative payoff at maturity t_N (depends on the coin tosses, e.g. $(K - S_N)^+$).
V_n	arbitrage-free derivative price at time t_n (at the current node).
$(1+r)^{N-n}$	discount over the *remaining* $N-n$ periods.
$\mathbb{E}^*_n$	risk-neutral expectation given what is known at t_n.
at $n = 0$	$V_0 = \mathbb{E}^*\left[V_N/(1+r)^N\right]$: price = discounted risk-neutral expected payoff.

- **Check on [S12] tree** *[Beyond slides]*: $M_0 = 4$; $\mathbb{E}^*[M_1] = (0.5 \cdot 8 + 0.5 \cdot 2)/1.25 = 4$ ✓.

- **Interpretation** *[Beyond slides]*: under $\mathbb{Q}$, every traded asset earns the risk-free rate on average, so discounting at r is correct. Under $\mathbb{P}$ the discounted stock drifts up (risk premium) and is not a martingale.
- **Traps:** claiming S_n (undiscounted) is a martingale when $r > 0$; using $\mathbb{P}$ in the pricing formula; discounting by $(1+r)^N$ instead of $(1+r)^{N-n}$ at time n.

1.11 Complete markets and arbitrage [S24–S25]

- **Complete market** [S24]: every derivative can be replicated from basic instruments (cash and the underlying), usually by dynamic rebalancing.
- **Incomplete market** [S24]: e.g. with trading frictions (or the trinomial model of S1), replication is impossible.
- **Two sides of one coin** [S24]: replicate the option from stock and cash, or hedge the option with stock to create cash.
- **Claims on** [S25]:
  - Absence of arbitrage implies the existence of risk-neutral probabilities.
  - Real world: riskier assets have higher expected returns. Risk-neutral world: all assets grow at the risk-free rate.
  - The denominator (numeraire) need not be the bank account; the choice is arbitrary.
- **The fundamental theorems** *[Beyond slides]*:
  1. No arbitrage $\iff$ there exists at least one equivalent martingale measure $\mathbb{Q}$.
  2. The market is complete $\iff$ that $\mathbb{Q}$ is unique.

  Binomial: unique $p^* \Rightarrow$ complete. Trinomial: a range of $p^* \Rightarrow$ incomplete, price interval (S1).
- △**!** [S25] "In a complete market …there is no arbitrage": completeness does not imply no-arbitrage; they are separate conditions (theorems 1 and 2 above).
- △**!** [S25] "all relative price processes should be a martingale": prices divided by the bank account are $\mathbb{Q}$-martingales. Dividing by a different asset (numeraire) gives martingales under a different measure associated with that numeraire *[Beyond slides]*. That is what "the choice is arbitrary" means:

$$\frac{V_n}{\mathcal{N}_n} = \mathbb{E}_n^{\mathbb{Q}^{\mathcal{N}}}\left[\frac{V_{n+1}}{\mathcal{N}_{n+1}}\right] \qquad \textit{[Beyond slides]}$$

where

$\mathcal{N}_n$	price of the chosen numeraire asset at t_n (must be strictly positive); the bank account $(1+r)^n$ is the default choice.
$V_n/\mathcal{N}_n$	relative price: the asset measured in units of the numeraire.
$\mathbb{Q}^{\mathcal{N}}$	the measure attached to numeraire $\mathcal{N}$ (equals $\mathbb{Q}$ when $\mathcal{N}_n = (1+r)^n$).

2 Comparison tables

2.1 Real-world $\mathbb{P}$ vs risk-neutral $\mathbb{Q}$

	$\mathbb{P}$ (empirical)	$\mathbb{Q}$ (risk-neutral)
Up probability [S4]	0.8 (a view)	0.5 (from no-arbitrage)
Expected stock growth	risk-free + risk premium	risk-free rate
$\mathbb{E}[S_2]$ in [S5]	112 (not enforceable)	100 (enforceable by borrow-and-buy)
Discounted price	not a martingale	martingale [S21]
Used for	forecasting, risk management	pricing [S23]
Link	$\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}[X\,\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}]$; requires equivalence	

2.2 Arithmetic vs geometric tree

	Arithmetic [S7–S9]	Geometric [S11, S14–S17]
Step	$S \pm \sigma\sqrt{T/n}$	$S \times u$ or $S \times d$; log step $(r - \frac{1}{2}\sigma^2)\frac{T}{n} \pm \sigma\sqrt{T/n}$
CLT applied to	sum of steps	sum of log steps
Limit	$S_T = S_0 + \sigma\sqrt{T}Z$ (normal)	$S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}Z}$ (lognormal)
Mean	S_0 $(r = 0)$	$S_0 e^{rT}$
Negative prices	possible	impossible
σ measured in	price units	log (percentage) units
Continuous-time analogue	$\mathrm{d}S = \sigma\,\mathrm{d}W$ (S4)	$\mathrm{d}S = rS\,\mathrm{d}t + \sigma S\,\mathrm{d}W$ (GBM)

2.3 European vs American on a tree

	European	American
Node value	$\frac{p^* V^u + q^* V^d}{1+r}$	max(continuation, exercise)
Closed form	$\mathbb{E}^*[V_N]/(1+r)^N$	none; backward induction only
[S12–S13] put	0.96	1.36 (exercise at $S = 2$)
Relation	$V^A \geq V^E$	premium = value of the early-exercise right

2.4 Conditional expectation rules [S20]

Linearity	$\mathbb{E}_n[aX + bY] = a\mathbb{E}_n[X] + b\mathbb{E}_n[Y]$
Known factor	$\mathbb{E}_n[XY] = X\mathbb{E}_n[Y]$ if X is known at n
Tower	$\mathbb{E}_n[\mathbb{E}_m[X]] = \mathbb{E}_n[X]$, $n \leq m$
Independence	$\mathbb{E}_n[X] = \mathbb{E}[X]$ if X is independent of the first n tosses

3 Master formula sheet

Each line repeats the key symbols so the sheet can be used on its own.

Change of measure [S2–S6]

- $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}(\omega) = \frac{\mathbb{Q}(\omega)}{\mathbb{P}(\omega)}$; equivalence $\mathbb{P}(\omega) > 0 \iff \mathbb{Q}(\omega) > 0$.
  where ω = one path; $\mathbb{P}(\omega), \mathbb{Q}(\omega)$ = its real-world / risk-neutral probability.
- $\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}\big[X\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\big]$; $\mathbb{E}^{\mathbb{P}}\big[\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\big] = 1$ *[Beyond slides].*
  where X = any payoff; $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ = path-by-path reweighting factor.

Trees [S7–S13]

- Arithmetic: step $\pm\sigma\sqrt{T/n}$; $S_T \to S_0 + \sigma\sqrt{T}Z$.
  where σ = price-unit volatility; T = maturity; n = steps; $Z \sim N(0,1)$.
- No arbitrage: $d < 1 + r < u$. $p^* = \frac{(1+r)-d}{u-d}$, $q^* = \frac{u-(1+r)}{u-d}$. CRR: $d = 1/u$ $(u = e^{\sigma\sqrt{\Delta t}}$ *[Beyond slides]*).
  where u, d = gross up/down factors; r = per-period rate; p^*, q^* = risk-neutral up/down probabilities; $\Delta t = T/N$.
- European: $V_n = \frac{p^* V^u_{n+1} + q^* V^d_{n+1}}{1+r}$; $V_0 = \frac{1}{(1+r)^N}\sum_j \binom{N}{j} p^{*j} q^{*N-j} V_N(j)$ *[Beyond slides].*
  where V^u_{n+1}, V^d_{n+1} = successor-node values; j = number of ups; $V_N(j)$ = payoff with j ups; N = periods.
- American put: $V_n = \max\big(\frac{p^* V^u + q^* V^d}{1+r},\ (K - S_n)^+\big)$.
  where first argument = continuation value; $(K - S_n)^+$ = exercise value; K = strike; S_n = stock at the node.
- Hedge *[Beyond slides]*: $\Delta_n = \frac{V^u - V^d}{S^u - S^d}$.
  where Δ_n = shares held at step n; S^u, S^d = next-step stock prices.

Lognormal limit [S14–S17]

- $S_T = S_0 \exp\big[(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z\big]$; $\mathbb{E}[\ln S_T] = \ln S_0 + (r - \frac{1}{2}\sigma^2)T$; $\mathbb{V}[\ln S_T] = \sigma^2 T$; $\mathbb{E}[S_T] = S_0 e^{rT}$.
  where r = continuously compounded rate; σ = log-volatility; $-\frac{1}{2}\sigma^2$ = convexity correction; $Z \sim N(0,1)$.

Martingales and pricing [S19–S23]

- $S_n = \frac{1}{1+r}\mathbb{E}^*_n[S_{n+1}]$; $p^*u + q^*d = 1 + r$.
  where $\mathbb{E}^*_n$ = risk-neutral expectation given the first n tosses.
- $M_n = \frac{S_n}{(1+r)^n} = \mathbb{E}^*_n[M_{n+1}]$; $M_0 = \mathbb{E}^*[M_N]$.
  where M_n = discounted stock; $(1+r)^n$ = bank account.
- $V_n = \mathbb{E}^*_n\left[\frac{V_N}{(1+r)^{N-n}}\right]$.
  where V_N = payoff at maturity; V_n = price at t_n; $N - n$ = remaining periods.

4 Glossary

- **American option:** exercisable at any time up to expiry [S13].
- **Backward induction:** rolling values from expiry to today node by node.
- **Central limit theorem:** standardised sums of i.i.d. variables tend to $N(0,1)$ [S8].
- **Complete market:** every payoff can be replicated [S24].
- **Conditional expectation $\mathbb{E}_n$:** expectation given the first n tosses [S19].
- **Continuation value:** discounted risk-neutral expectation of next-step values; what holding is worth [S13].
- **Cox–Ross–Rubinstein (CRR):** geometric tree with $d = 1/u$ [S11].
- **Early-exercise premium:** $V^A - V^E$.
- **Equivalent measures:** agree on which events have zero probability [S3].
- **Intrinsic (exercise) value:** payoff from exercising now, e.g. $(K - S_n)^+$.
- **Martingale:** $M_n = \mathbb{E}_n[M_{n+1}]$; best forecast of tomorrow is today [S21].
- **Numeraire:** the asset prices are divided by (bank account by default) [S25].
- **Probability space:** sample space plus probability measure [S2].
- **Radon–Nikodym derivative:** ratio $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ reweighting outcomes [S2].
- **Risk-neutral measure $\mathbb{Q}$:** measure under which discounted prices are martingales [S21–S23].
- **Risk-neutral pricing formula:** $V_n = \mathbb{E}^*_n[V_N/(1+r)^{N-n}]$ [S23].
- **Sample space Ω:** set of all outcomes (paths) [S2].
- **Tower property:** $\mathbb{E}_n[\mathbb{E}_m[X]] = \mathbb{E}_n[X]$ for $n \le m$ [S20].

5 Self-test

Questions

Q1. In the [S4] example, compute $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ on each two-step path and verify that $\mathbb{E}^{\mathbb{P}}[S_2\,\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}] = 100$ and $\mathbb{E}^{\mathbb{P}}[\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}] = 1$.

Q2 (conceptual). What does it mean for $\mathbb{P}$ and $\mathbb{Q}$ to be equivalent, and why is it needed?

Q3. In an arithmetic tree with n steps and $r = 0$, why is each step $\pm\sigma\sqrt{T/n}$? Show $S_T = S_0 + \sigma\sqrt{T}Z$ has the required mean and variance.

Q4. $S_0 = 50$, $u = 1.2$, $d = 0.9$, $r = 5\%$ per period. Check no-arbitrage, find p^*, and price a one-period call with $K = 52$. Give the hedge.

Q5 ([S12]). $S_0 = 4$, $u = 2$, $d = \frac{1}{2}$, $r = 25\%$, $K = 5$, two periods. Price the European put and call; check put–call parity.

Q6 ([S13]). Price the American put on the same tree. Where is it optimal to exercise?

Q7 ([S16–S17]). For $S_T = S_0 \exp[(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z]$, find $\mathbb{E}[\ln S_T]$, $\mathbb{V}[\ln S_T]$ and $\mathbb{E}[S_T]$. Why the $-\frac{1}{2}\sigma^2$?

Q8 ([S18]). Three tosses, X = heads, Y = tails. Give $X(\omega), Y(\omega)$ for all outcomes, and the distributions under $p = \frac{1}{2}$ and $p = \frac{2}{3}$.

Q9 ([S22]). Prove that $S_n/(1+r)^n$ is a $\mathbb{Q}$-martingale, naming each conditional-expectation rule you use.

Q10 (conceptual). State the two fundamental theorems of asset pricing and use them to classify the binomial and trinomial models. What is wrong with "complete market ⇒ no arbitrage"?

Solutions

S1. Paths: UU 25/64, UD and DU 25/16, DD 25/4. $0.64(120)(25/64) + 0.32(100)(25/16) + 0.04(80)(25/4) = 30 + 50 + 20 = 100$. $0.64(25/64) + 0.32(25/16) + 0.04(25/4) = 0.25 + 0.5 + 0.25 = 1$.

S2. They agree on which outcomes are possible: $\mathbb{P}(\omega) > 0 \iff \mathbb{Q}(\omega) > 0$. Otherwise the ratio is undefined somewhere and information about one measure is lost; economically, an arbitrage under one measure would not be one under the other.

S3. Each step needs variance $\sigma^2 T/n$ so that n independent steps sum to $\sigma^2 T$; a $\pm a$ step with probability $\frac{1}{2}$ has variance a^2, so $a = \sigma\sqrt{T/n}$. $\mathbb{E}[S_T] = S_0$, $\mathbb{V}[S_T] = \sigma^2 T\, \mathbb{V}[Z] = \sigma^2 T$.

S4. $0.9 < 1.05 < 1.2$ ✓. $p^* = (1.05 - 0.9)/0.3 = 0.5$. Payoffs: $60 - 52 = 8$, $(45 - 52)^+ = 0$. $C_0 = 0.5 \cdot 8/1.05 = 3.81$. Hedge: $\Delta = 8/15 = 0.533$ shares, bond $= C_0 - \Delta S_0 = 3.81 - 26.67 = -22.86$ (borrow 22.86).

S5. $p^* = 0.5$. Put: payoffs $0, 1, 4 \Rightarrow 0.40, 2.00 \Rightarrow 0.96$. Call: payoffs $11, 0, 0 \Rightarrow 0.25 \cdot 11/1.5625 = 1.76$. Parity: $1.76 - 0.96 = 0.80 = 4 - 5/1.5625$ ✓.

S6. Node $S = 2$: $\max(2.00, 3) = 3$ (exercise). Node $S = 8$: $\max(0.40, 0) = 0.40$. Today: $\max((0.2 + 1.5)/1.25, 1) = \max(1.36, 1) = 1.36$. Exercise at $t = 1$ after a down move.

S7. $\mathbb{E}[\ln S_T] = \ln S_0 + (r - \frac{1}{2}\sigma^2)T$; $\mathbb{V} = \sigma^2 T$; $\mathbb{E}[S_T] = S_0 e^{(r-\frac{1}{2}\sigma^2)T} e^{\frac{1}{2}\sigma^2 T} = S_0 e^{rT}$. The $-\frac{1}{2}\sigma^2$ offsets the lognormal convexity so the stock grows at r under $\mathbb{Q}$.

S8. HHH (3,0); HHT, HTH, THH (2,1); HTT, THT, TTH (1,2); TTT (0,3). Fair: both Bin$(3, \frac{1}{2})$: $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$. $p = \frac{2}{3}$: X: $\frac{1}{27}, \frac{6}{27}, \frac{12}{27}, \frac{8}{27}$ for 0..3; Y reversed.

S9. $\mathbb{E}_n^*[S_{n+1}/(1+r)^{n+1}] = \frac{S_n}{(1+r)^{n+1}} \mathbb{E}_n^*[S_{n+1}/S_n]$ (known factor S_n) $= \frac{S_n}{(1+r)^{n+1}} \mathbb{E}^*[S_{n+1}/S_n]$ (independence: depends only on toss $n+1$) $= \frac{S_n}{(1+r)^{n+1}}(p^*u + q^*d) = \frac{S_n}{(1+r)^n}$ since $p^*u + q^*d = 1 + r$.

S10. (1) No arbitrage $\iff$ an equivalent martingale measure exists. (2) Complete $\iff$ it is unique. Binomial with $d < 1 + r < u$: unique $p^* \Rightarrow$ arbitrage-free and complete. Trinomial: many $p^* \Rightarrow$ arbitrage-free but incomplete. A complete market can still contain an arbitrage: take a binomial tree with $u < 1 + r$. Every payoff is still replicable (two states, two assets), but shorting the stock and lending the proceeds wins in every state, and no $p^* \in (0, 1)$ exists.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title	header
S2	Probability space; changing measure; Radon–Nikodym derivative	1.1
S3	When the derivative exists; equivalence of measures	1.1
S4	Example: fund manager (0.8) vs market maker (0.5); $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ tree	1.2
S5	$\mathbb{E}^{\mathbb{P}}[S_2] = 112$, $\mathbb{E}^{\mathbb{Q}}[S_2] = 100$; change-of-measure formulas	1.2
S6	Verification of both formulas	1.2
S7	Arithmetic binomial tree construction	1.3
S8	Central limit theorem; normal limit	1.3
S9	Example: mean and variance of $S_0 + \sigma\sqrt{T}Z$	1.3
S10	Non-zero rates; $S^d < S_0(1+r) < S^u$; p^*	1.4
S11	Geometric tree; u, d; p^*; CRR $d = 1/u$	1.5
S12	Example: European put on a 2-step tree	1.5
S13	American options; example	1.6
S14	CLT for products ⇒ lognormal	1.7
S15	Log-price tree with drift $r - \frac{1}{2}\sigma^2$	1.7
S16	Lognormal limit; example on $\mathbb{E}[\ln S_T]$, $\mathbb{V}[\ln S_T]$	1.7
S17	Example: $\mathbb{E}[S_T] = S_0 e^{rT}$ via MGF	1.7
S18	Coin example; random variables vs measure; finite expectation	1.8
S19	Conditional expectation; $S_n = \mathbb{E}_n^*[S_{n+1}]/(1+r)$	1.9
S20	Properties of conditional expectation	1.9
S21	Discounted stock is a martingale; $M_0 = \mathbb{E}^*[M_N]$	1.10

Slide	Content	Section
S22	Proof of the martingale property	1.10
S23	Risk-neutral pricing formula	1.10
S24	Complete vs incomplete markets; replication	1.11
S25	No arbitrage $\Rightarrow \mathbb{Q}$; risk-free growth; numeraires	1.11

Appendix B. Flag summary (△!)

1. [S2] Probability space is $(\Omega, \mathcal{F}, \mathbb{P})$; $\mathcal{F}$ omitted because Ω is finite.
2. [S3] With $\mathbb{P}(\omega) = 0 < \mathbb{Q}(\omega)$ it is $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ that is undefined, not $\mathbb{P}/\mathbb{Q}$ (which is 0).
3. [S10] vs [S15–S17]: r is a per-period simple rate in the tree slides but continuously compounded in the lognormal-limit slides.
4. [S11] $u = S_1(H)/S_0$ uses coin notation (H/T) introduced only on [S18]; the T in $S_1(T)$ means tails, not maturity.
5. [S15] The $\pm\sigma\sqrt{T/n}$ log tree is risk-neutral only in the limit $n \to \infty$.
6. [S25] "Complete market $\Rightarrow$ no arbitrage" is not a theorem; no-arbitrage $\iff$ existence of $\mathbb{Q}$, completeness $\iff$ uniqueness.
7. [S25] Relative prices are $\mathbb{Q}$-martingales only with the bank account as denominator; another numeraire comes with its own measure.
8. [S9, S12, S13, S16, S17] Examples are posed without worked answers; the solutions here are my own.